Project Design Phase-II: Data Flow Diagram & User Stories

Global Olympic Sports Performance Analysis (1992–2026)

Date: 31 January 2026	**Team ID:** NOC-PERF-2026
Project Name: Global Olympic Sports Performance Analysis	**Maximum Marks:** 4 Marks

Data Flow Diagram (DFD) Architecture & Symbols

Symbol	Name	Description & Project Context
Oval / Rounded	External Entity	Stakeholders initiating or receiving system outputs (e.g., Ms. Priya / NOC Strategist, Mr. Carlos / Funding Agency, William Robert / Data Journalist).
Rectangle (Numbered)	Process	Data transformation activities (e.g., 1.0 Data Ingestion & Validation, 2.0 Calculate Per-Athlete Medal Yield, 3.0 Generate Tableau Analytics).
Rectangle (Solid Fill)	Data Store	Repositories holding structured Olympic data (e.g., DS-1 Athletes Dataset, DS-2 Medal Results Dataset, DS-3 Financial & Regional Investment Store).
Labeled Arrow	Data Flow	Movement of data between entities, processes, and stores (e.g., Raw Medal Data, Aggregated Efficiency Metrics, Dashboard Visual Filters).

User Stories Matrix

User Type	Functional Epic	USN	User Story / Task	Acceptance Criteria	Priority	Release
NOC Strategist (Ms. Priya)	Per-Athlete Medal Yield	**USN-1**	As an NOC Strategist, I want to view per-athlete medal yield by sport category so that I can invest in high-density sports like Rowing (6.92) and Shooting (6.73).	Filter bar chart by sport category and rank top 10 per-athlete yields.	**High**	Sprint-1
NOC Strategist (Ms. Priya)	Country Benchmarking	**USN-2**	As an NOC Strategist, I want to benchmark our total medals against leaders like USA (735) and China (696) to identify competitive gaps.	Interactive country comparison view with historical side-by-side totals.	**High**	Sprint-1
Investment Analyst (Mr. Carlos)	Investment Efficiency	**USN-3**	As a Funding Analyst, I want to plot national sports investment against medal counts in a bubble chart to spot overperforming nations like Slovakia and Ethiopia.	Bubble size maps to efficiency ratio; color maps to geographic region.	**High**	Sprint-2
Investment Analyst (Mr. Carlos)	Regional Reallocation	**USN-4**	As a Funding Analyst, I want to evaluate regional efficiency percentages (South America at 49.64%) via a pie chart to guide grant allocations.	Pie chart calculates regional efficiency % based on filtered historical data.	**Medium**	Sprint-2
Broadcast Journalist (William Robert)	Host Advantage Tracking	**USN-5**	As a Broadcast Journalist, I want to compare host nation medal tallies (Japan: 105, USA: 100, China: 93) to highlight home advantage narratives.	Multi-bar host nation dashboard with historical comparisons across Games.	**High**	Sprint-1
Broadcast Journalist (William Robert)	Historical Trend Line	**USN-6**	As a Broadcast Journalist, I want to trace medal totals over time (305 in 2000 to 363 in 2020) to visualize Olympic expansion.	Line chart displaying total medals awarded per year from 1992 to 2026.	**Medium**	Sprint-2